Project Design Phase
Solution Architecture

Date	02 October 2026
Team ID	SWTID-2026-5435
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram: Automating Standard Laptop Orders with Flow Designer

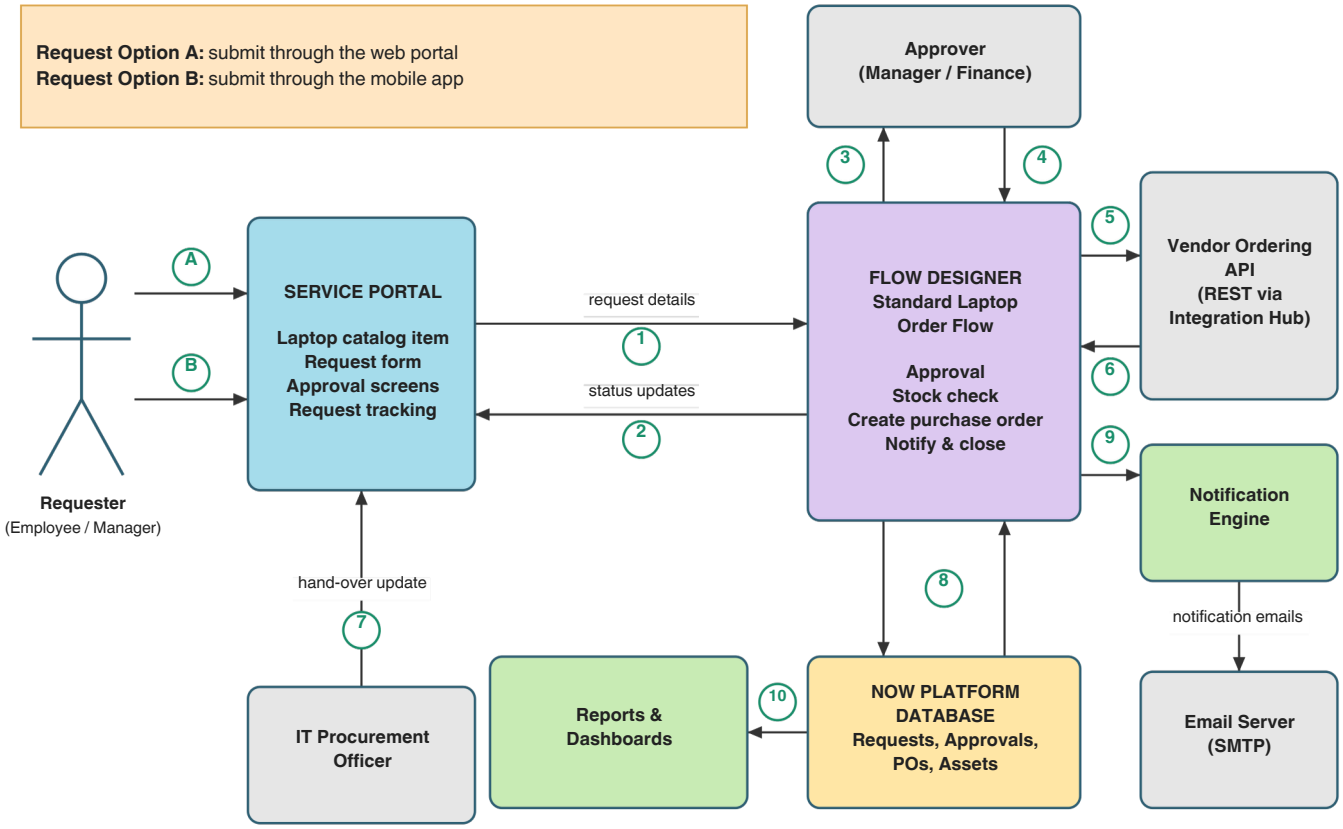


Figure 1: Architecture and data flow of the automated standard laptop ordering solution

Data Flow Description

The numbers below match the numbered flows in Figure 1.

A / BThe requester submits the standard laptop catalog item through the web portal (Option A) or the mobile app (Option B).

1The portal sends the request details (employee, role, location, joining date, laptop model) to the Standard Laptop Order flow in Flow Designer.

2The flow sends status updates back to the portal so the requester can track the request.

3The flow sends an approval request to the manager / budget approver.

4The approver's decision (approve or reject, with comments) returns to the flow. Reminders and escalation apply if there is no response.

5After approval and a stock check, the flow creates a purchase order and sends it to the vendor through a REST API (Integration Hub).

6The vendor returns shipment and invoice details, which the flow attaches to the request.

7The IT procurement officer records receipt and hand-over of the laptop through the portal, and the update goes to the flow.

8Requests, approvals, purchase orders and asset records are read from and written to the Now Platform database.

9The notification engine sends emails through the email server to the requester and approver at every status change.

10Stored data feeds reports and dashboards for approval time, order volume and delivery time.

Reference:

https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/